



Protocol name: Post-staining easyIRF sections for electron microscopy

Version: 1.1

Status: Draft

Date Approved: 03/04/2024

Author: Dumisile Lumkwana/ Lucy Collinson

Keywords: EasyIRF; Cell; Section; Uranyl Acetate; Potassium Permanganate; Lead Citrate

Abbreviations: IRF - In Resin Fluorescence; UA - Uranyl Acetate; ITO - Indium Tin Oxide, Potassium Permanganate (KMnO_4)

Health & Safety requirements

Ensure you are familiar with all relevant local Health and Safety documents and procedures (Material Safety Data Sheets, Procedural Risk Assessments, Standard Operating Procedures etc.) before starting.

Safety critical: All procedures designated safety critical must be performed in a fume hood (where appropriate) and the following PPE must be worn - lab coat, gloves, and eye protection.

Procedural Risk Assessments: Handling of uranyl acetate, lead citrate, potassium permanganate, sodium hydroxide

Materials

- ☐ 15 ml glass scintillation vials with caps
- ☐ 2 ml cryovial
- ☐ Hot plate set to 60°C
- ☐ Reverse fine tip tweezers
- ☐ Glass slide
- ☐ Plastic box with lid
- ☐ Parafilm
- ☐ Glass beakers
- ☐ Aluminium foil
- ☐ 20 ml syringes
- ☐ 22 μm filter
- ☐ Blotting paper
- ☐ Aluminium pin SEM stubs (check compatibility with microscope) Agar AGG301
- ☐ Sticky carbon pad Agar AGG3347N
- ☐ Silver paint Agar AGG3691

- ☐ Roller mixer
- ☐ Vortex

Chemicals

- ☐ ddH₂O; double distilled water
- ☐ Sodium hydroxide pellets

Reagents

- ☐ Staining solution 1 - 2% uranyl acetate + 0.5% potassium permanganate
- ☐ Staining solution 2 - Reynold's lead citrate

Staining solution 1

Stock solution: 4% uranyl acetate **Safety critical**

Use **Reagent Protocol RP01: Preparation of aqueous uranyl acetate**, adjusting the concentration to make 4% uranyl acetate.

Uranyl Acetate	State: Solid	Hazard Codes: H300, H330, H373, H411	
----------------	-----------------	--	--

Stock solution: 1% potassium permanganate

Use **Reagent Protocol RP03: Preparation of potassium permanganate**

Potassium Permanganate	State: Solid	Hazard codes: H272, H302, H314, H318, H361d, H400, H410	
------------------------	-----------------	--	--

Working solution: 2% uranyl acetate + 0.5% potassium permanganate **Safety critical**

Each sample (ITO) you wish to stain;

Add 500 ul of 4% uranyl acetate to a 2 ml cryovial.

Freeze at -20°C.

Add 500 ul of 1% KMnO₄ to the frozen uranyl acetate and thaw at RT.

Once thawed, vortex the mixture.

Staining solution 2

Use **Reagent Protocol RP02: Preparation of Reynold's Lead Citrate** **Safety critical**

Lead Citrate	State: Liquid	Hazard Codes: H302, H332, H360Df, H373, H410	
--------------	------------------	--	--

Procedure

1. Prepare Staining Solution 1 and Staining Solution 2. **Safety critical**
2. Set and heat the hotplate to 60°C.
3. Label the tweezers and a clean glass slide with the sample name.
4. Place the ITO coverslip with the sections facing upwards on the glass slide and then place the slide onto the hotplate for 10 mins.

Note: This baking step prevents the sections from detaching from the ITO coverslip during the staining and washing steps.

5. Line the lid of the plastic box with parafilm and place one drop of Staining Solution 1 (~ 1 mL) using a syringe with filter and one drop of ddH₂O (~ 1 mL) onto the parafilm for each coverslip to be stained (Fig.1A). **Safety critical**

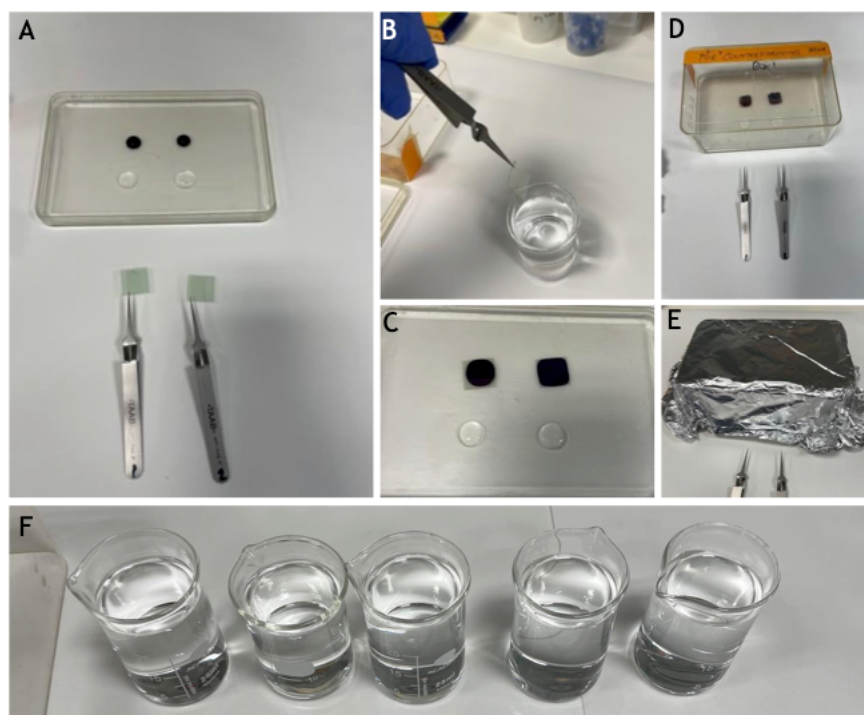


Figure 1. Layout for staining solution 1 and washing of ITO coverslips.

6. Wet the ITO coverslip by dipping it in a glass beaker of ddH₂O (Fig.1B).
7. Place the ITO coverslip onto the drop of stain with the sections facing down (the ITO should float on the drop). Repeat step 6 and 7 for the remaining ITO coverslips (Fig.1C).

8. Cover with lid (Fig.1D).
9. Cover the lid with aluminium foil (Fig.1E).
10. Leave ITO coverslip(s) on the stain for 10 mins.
11. Fill 5 glass beakers with ddH₂O (Fig.1F).
12. After 10 mins, carefully lift the ITO coverslip and place onto the drop of water. Repeat for remaining ITO coverslips.
13. Carefully lift the ITO coverslip from the drop of water and rinse by dipping ten times vertically in the first glass beaker of ddH₂O, then repeat for the remaining four beakers to complete 50 x fast washes. Repeat this step for the remaining ITO coverslips.
14. Blot the coverslips dry at the edge, being careful not to touch the sections or deposit lint onto the glass.

Note: Dispose of mixed uranyl acetate and potassium permanganate waste according to local rules.

Note: If sections are lifting off after lead citrate staining, first reduce the washes to 3x beakers 10 dips instead of 5x, if this does not help then reduce the post staining with lead citrate to 2 mins.

15. Exchange the parafilm lining the lid and add sodium hydroxide pellets close to where the staining droplet will be to absorb carbon dioxide and prevent precipitation of the lead (Fig.2A). **Safety critical**

Sodium Hydroxide	State:	Hazard Codes:	
	Solid	H290, H314	

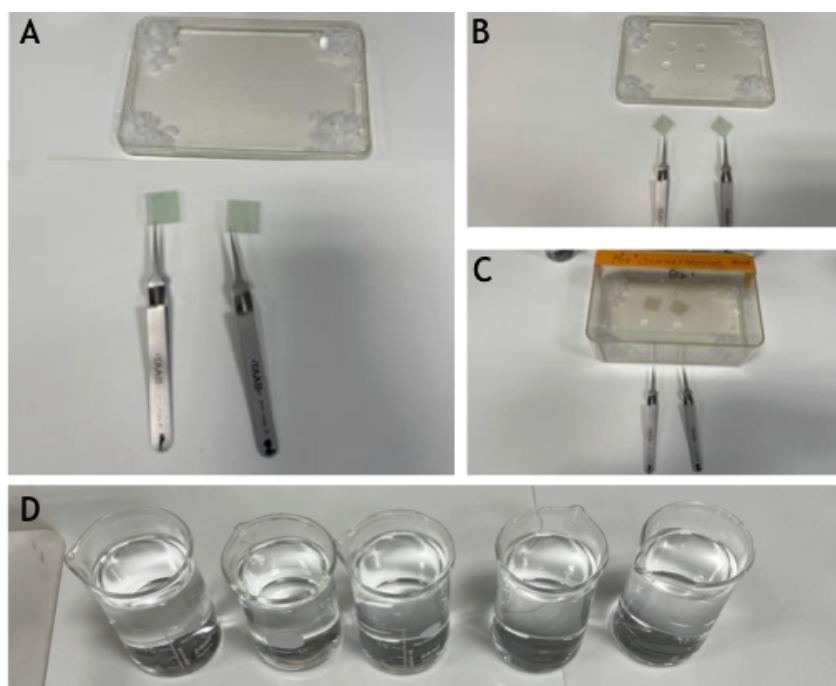


Figure 2. Layout for staining solution 2 and washing of ITO coverslips.

16. Place one drop of Staining Solution 2 (~ 1 mL) using a syringe with filter and one drop of ddH₂O (~ 1 mL) onto the parafilm for each coverslip to be stained (Fig.2B).

Note: Work quickly and minimize exposure of the stain drops to carbon dioxide. Do not breathe onto the drops.

17. Place the ITO coverslip onto the drop of stain with the sections facing down (the ITO should float on the drop) and immediately cover with lid. Repeat step 16 and 17 for the remaining ITO coverslips (Fig.2C).
18. Leave ITO coverslip(s) on the stain for 10 mins.
19. Fill 5 glass beakers with ddH₂O (Fig.2D).
20. Repeat step 12 to 14
21. Leave the coverslips to dry in the reverse forceps overnight, protected from dust and in the dark.
22. Put the silver paint bottle on a roller mixer to resuspend for 5 mins.

Quick Drying Silver Paint	State: Liquid	Hazard Codes: H225, H302, H319, H335	 
---------------------------	-------------------------	---	--

23. Place an adhesive carbon tab on an SEM stub and place the ITO coverslip on it with the sections facing up (Fig.3).

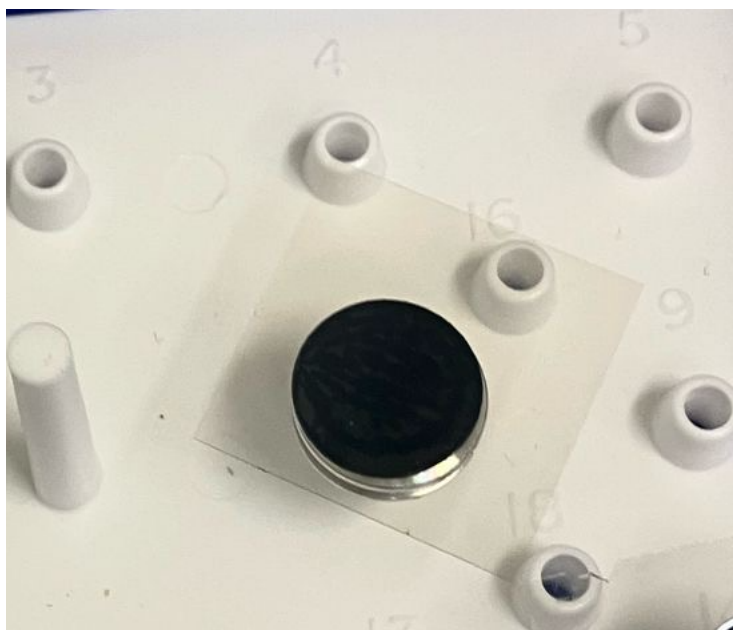


Figure 3. Mounting coverslips on SEM stubs.

24. Paint part of the edge of the coverslip and stub with silver paint so there is a conductive path from the glass to the metal of the stub (Fig.4).

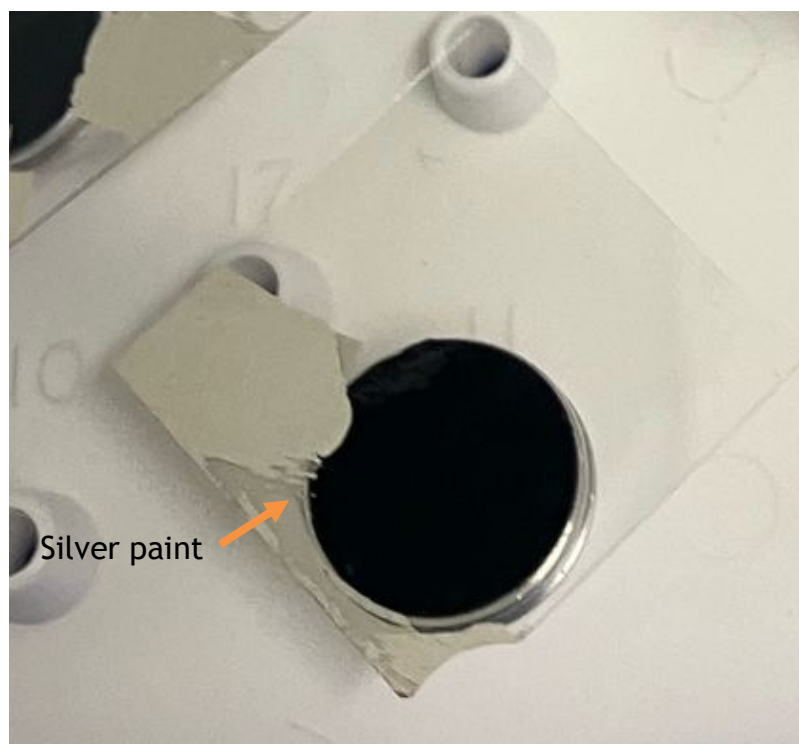


Figure 4. Paint stub and ITO with silver paint.